

AREE

Autonomous Regulatory Escalation Engine

End-to-End Engineering Report

SIH PS 26082 — Air Pollution–Weather Coupled Forecasting for Delhi NCR
Ministry of Earth Sciences / NCMRWF · Disaster Management
Team Devengers

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1. Executive summary

AREE forecasts whether the atmosphere over Delhi NCR will remain able to clear itself, converts that into an early warning with a stated lead time, and connects it to deterministic regulatory rules that a human authorises. It is not an AQI dashboard: the product is **lead time**, not a concentration number.

This report documents the whole programme end to end — what was built, what was measured, what failed and was withdrawn, the architecture as it now stands, the database that backs it, and the route to a WRF-Chem coupled core. It is deliberately written to be checkable: every experiment states its protocol, and every negative result is kept.

The single most important finding

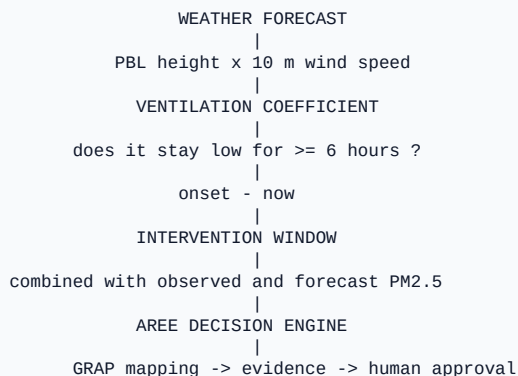
Judged on mean absolute error, climatology is a competitive baseline and our model looked mediocre. Judged on **warning skill** — did you call the episode before it started, and how early — climatology scores **zero**, persistence scores **zero**, and the upper-tail model anticipates **9 of 13** severe episodes with a median **68 hours** of lead. The choice of metric completely inverts the ranking. That is the core scientific argument of the submission.

Status at a glance

Component	Status	Evidence
Feature store (5 tables + target)	Built	152.9 MB, 9 met grid points
Baselines (persistence, climatology)	Built + scored	Walk-forward, Nov 2021–2024
lgbm-v1 central forecast	Built	1 clean win of 3 comparable folds
q90 upper-tail risk	Built + validated	9/13 episodes, 68 h median lead
Experiment B (operational skill)	Done	Weather-forecast cost 0–3 MAE
Experiment C (failure diagnosis)	Done	Cause was our objective, not physics
Experiment D (warning skill)	Done	Baselines have 0 cold-start skill
Decision engine + GRAP + approval	Built	AWAITING_APPROVAL, human authorises
/api/aree/outlook vertical slice	Built	HTTP → UI → PDF verified
Live multi-station capture	Running	Hourly, Windows Task Scheduler
λ aerosol–PBL feedback index	WITHDRAWN	AUC 0.504 on real data — chance
WRF-Chem coupled core	Not built	Interface defined; see section 11

2. The problem, and the shape of the answer

A conventional monitoring dashboard tells an authority “PM2.5 is 250”. That is a measurement of the present. The operational questions are different: what will the atmosphere do next, will it become unable to disperse pollution, when does that begin, and how much time remains to act?



Ventilation coefficient = boundary-layer height x 10 m wind speed, in m²/s. Mixing depth times transport speed: the volume flux available to dilute whatever is emitted. It is a long-standing quantity in air-quality meteorology. We did not invent it; what is ours is the operational chain built on top of it.

Delhi's winter smog is not primarily an emissions story. It is this curve failing to rise: solar heating grows the mixed layer through the morning so dispersion capacity rises by two orders of magnitude, then collapses after sunset.

3. Data layer — what enters the system

Every source below is a real external feed. Each is used for what it is actually authoritative on, and every record states which source it came from.

Source	Supplies	Why this one	Status
Open-Meteo	PBLH, wind, radiation, T, RH	ECMWF-family fields over plain REST, no key, whole 72 h horizon	Working
CAQM (caqm.nic.in)	Live NCR roster + sub-index AQI	79 min median behind vs data.gov.in at 322 min	Working
CPCB via data.gov.in	PM2.5 concentrations (µg/m ³)	CAQM publishes sub-indices only; the episode threshold is calibrated	Working
OpenAQ v3	Station provenance, recent hourly readings	Independent cross-check on the network	Working
NASA FIRMS	Fire detections, FRP	Crop-residue transport attribution	Working
WAQI	Pan-India AQI	Coverage outside NCR	Unreliable — do not claim live

3.1 The live weather call

```
https://api.open-meteo.com/v1/forecast
?latitude=28.63&longitude=77.22
&hourly=boundary_layer_height,shortwave_radiation,terrestrial_radiation,
cloud_cover,temperature_2m,relative_humidity_2m,wind_speed_10m,
wind_direction_10m,surface_pressure,precipitation
&forecast_days=3&timezone=UTC&wind_speed_unit=ms

PBLH x wind_speed_10m from this response IS the ventilation coefficient.
Verified against the raw payload: chart peak 1262.2 m2/s = 935 m x 1.35 m/s.
```

3.2 Different pages read different sources — deliberately

This is worth stating plainly because it is visible in the product. The national map and command centre read **CAQM** (73 stations, AQI sub-index, fresher). The ventilation and outlook views read **CPCB via data.gov.in** (80 stations, PM2.5 µg/m³, slower). CAQM carries no PM2.5 concentration anywhere in its payload, and inverting the CPCB breakpoint table to manufacture one would fabricate precision inside the number the whole decision turns on. So the station counts differ, and that is correct rather than a bug.

4. What failed — kept, not buried

These are recorded because a reviewer will ask, and because each one changed the design.

4.1 λ — the aerosol-PBL feedback index. WITHDRAWN.

The original scientific claim was that a measurable feedback-loop gain λ would separate pollution episodes that lock in from those that ventilate out. The estimator was validated on synthetic data with a known λ and recovered it to 9.2% error, with all three physical sign checks passing. On **real** data it achieved **AUC 0.504** — chance. A PBL-free alternative reached **0.554** — also chance. Both claims were withdrawn, and no part of the shipped system depends on either.

4.2 What replaced it

Ventilation measured over the 48 hours *after* episode onset separates locked-in from ventilated episodes at **AUC 0.736**. That reframed the problem from diagnosis to forecasting, which is what the system now does.

Predictor	AUC	Verdict
λ — aerosol-PBL feedback gain	0.504	chance — withdrawn
PBL-free ventilation-failure index	0.554	chance — withdrawn
Ventilation measured before onset	0.514	chance
Ventilation over the 48 h after onset	0.736	informative — this is what we ship

4.3 The silent-failure family

Six separate defects in this programme shared one signature: an endpoint or a pipeline that **answered confidently rather than refusing**. Each produced a plausible wrong answer that looked exactly like a right one.

Where	What it did	How it was caught
OpenAQ /parameters/latest	Ignored bbox; returned global rows	'Delhi composite' built from Korean monitors
CAQM GetGoogleMapData	Served AQI with no timestamp	A 24-day-old reading looked live
Open-Meteo ERA5 archive	HTTP 200, all pressure levels null	Inversion column silently empty
Met grid backfill	4 of 9 grid points empty, reported success	Row count looked plausible
capture.py	Reported 79 stations, wrote 0 rows	Wrong value key (pm25 vs pm25_ugm3)
History audit	Uniform 0/720 looked like a broken query	Sensors were retired; sensor dates $\neq$ location dates

The rule adopted as a result: a zero-result dataset is not evidence until the query and the data path have themselves been validated. Every ingestion module now reports expected-versus-actual coverage rather than a bare row count.

5. Database and storage

A single SQLite file with portable SQL. PostgreSQL is the destination but was the wrong starting point: putting a server install between four people and the first joined dataset buys nothing. Every write is `INSERT ... ON CONFLICT DO UPDATE` on a natural key, so re-running any stage is idempotent and the move to PostgreSQL is a connection swap.

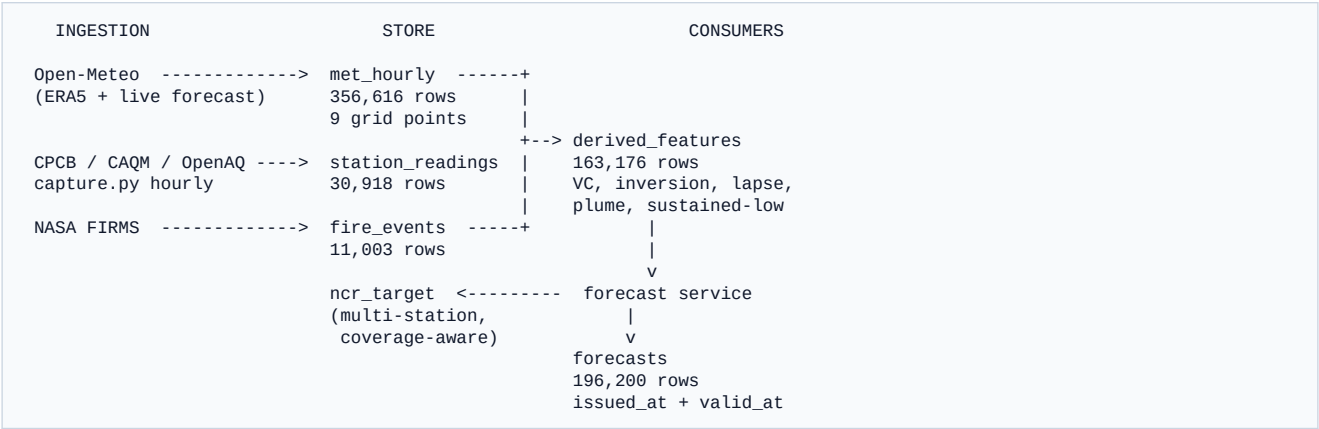


Table	Rows	Span	Purpose
station_readings	30,918	2020-10-01 .. 2026-09-02	Ground truth, one row per station-hour
met_hourly	356,616	2020-10-01 .. 2026-09-01	Meteorology per grid-hour
fire_events	11,003	2024-11-01 .. 2024-11-30	Satellite fire detections
derived_features	163,176	2020-10-01 .. 2026-09-01	Everything computed, rebuildable
forecasts	196,200	2021-11-01 .. 2024-11-30	Predictions, with BOTH timestamps
ncr_target	1	2026-09-02 .. 2026-09-02	Multi-station NCR target (accumulating)

5.1 Why forecasts carry two timestamps

`issued_at` is what makes the table worth having. Without it there is no way to ask “what did we know, and when”, and a scoring run silently becomes a hindcast. Every forecast row records the moment it was made and the moment it is valid for, so skill can be measured at the lead time we would actually have had.

5.2 Ground-truth provenance

Source tag	Rows	Stations	What it is
research:ground_pm25_hourly	29,953	1	Historical NCR composite — 67% of hours rest on ONE monitor
openaq:hourly	486	7	Recent hourly history retrieved to unblock live lags
live:CPCB CAAQMS via data.gov.in	479	80	Hourly multi-station capture, running now

6. Experiments and results

6.1 Baselines first — the bar any model must clear

Walk-forward across four Novembers, each forecast using only data from before that season. Issue times every 6 h.

Lead	Persistence	Climatology	Winner
1–6 h	49.7	85.6	persistence
7–12 h	91.8	85.8	climatology
13–24 h	94.3	86.3	climatology
25–48 h	108.2	87.0	climatology
49–72 h	118.3	85.6	climatology

PM2.5 MAE $\mu\text{g}/\text{m}^3$, n-weighted. Persistence decays monotonically as its anchor goes stale; climatology is flat because it has no lead dependence. The crossover sits between 6 and 12 hours.

A methodology bug this caught. The first run issued once a day at 00:00 UTC, and persistence showed a violent spike at 7–12 h (bias +86 to +103) in all four winters. That was not a property of persistence: issuing at a fixed hour means “7–12 h ahead” always lands at the same time of day, so lead-time buckets were confounded with diurnal phase. Any future evaluation must vary the issue hour or it measures the diurnal cycle and calls it forecast skill.

6.2 lgbm-v1 — and the instability the pooled number hid

November	LGBM	Climatology	Verdict
2021	62.9 – 75.0	87.7 – 93.7	wins everywhere
2022	66.9 – 78.0	68.5 – 71.8	wins 1–6 h only
2023	59.8 – 86.7	90.8 – 95.4	wins everywhere
2024	90.1 – 98.9	86.2 – 88.5	LOSES everywhere

Pooled, the model looked 8–11% better than the best baseline. Per year, that was carried entirely by 2021 and 2023. In 2024 — the most recent and most severe November — it lost at every horizon. Reporting the pooled figure alone would have overstated what we have.

6.3 Experiment B — what does not knowing the weather cost?

The same fixed model driven by real past NWP forecasts instead of reanalysis. Cost: **0–3 MAE**. Set against the model's ± 20 MAE swing between years, the conclusion is blunt — **weather-forecast error is not the bottleneck; year-to-year generalisation is**. That redirected the research away from the meteorological inputs.

Bounded rather than exact: the previous-runs endpoint serves no forecast boundary-layer height (0/168 against 168/168 for the current run), and wind forecasts do not exist before 2024, so Experiment B runs on Nov 2024 only.

7. Diagnosing the failure — it was us, not the atmosphere

7.1 C0 — the four folds are not the same measurement

November	Hours	Coverage	Monitors (median)	% single
2021	651	90%	32	4%
2022	502	70%	1	100%
2023	522	72%	1	100%
2024	710	99%	1	100%

By season the switch is unmistakable — median monitors 15 → 21 → **1, 1, 1**, exactly at OpenAQ's Nov 2022 gap. Nov 2021, the model's most convincing win, is scored against a 32-station airshed average that no longer exists. On the mutually comparable folds the record is **one clean win of three**, not two of four.

7.2 C1 — the meteorology barely differs

November	VC p50	% below 466	Wind p50	PBLH p50
2023	100	71%	1.90	50
2024	144	69%	1.39	105

Near-identical dispersion regimes, opposite model verdicts. “2024 was a different atmospheric regime” is **falsified**. What differs is the target's right tail: three times as many hours above 400 µg/m³. In 2024, **68% of the model's error is one-directional under-forecast**, entirely inside the 400+ bin, and 47% of the year's total error comes from three days.

7.3 C2 — the cause was the loss function

November	Training max	Test max	Model's max prediction	Hours above ceiling
2023	926	599	413	4%
2024	926	1000	341	17%

The model never predicts above 341 µg/m³ although its training data reached 926. That is not extrapolation failure — it is the **L1 objective**, which is minimised by the conditional median and therefore compresses a right-skewed target toward its middle. Seventeen percent of Nov 2024 was unreachable by construction.

7.4 The objective experiment — everything else frozen

Variant	Overall MAE	400+ MAE	Max prediction
climatology	84.4	330.8	247
lgbm-v1 (L1, median)	83.9	337.7	413
lgbm-log-l1	83.4	350.7	460
lgbm-log-l2	88.6	355.2	473
lgbm-l2 (mean)	95.8	332.8	647
lgbm-q90 (upper tail)	110.2	266.8	714

Observed p99 across these folds is **629**. Only q90 produces a ceiling above it. The compression is real and fixable by the objective alone — but raising the tail costs overall accuracy. **No single objective serves both jobs**, which is the empirical case for a two-output forecast rather than an aesthetic preference.

8. Experiment D — the metric that inverts the ranking

MAE is not the currency of disaster management. An authority acts on “a severe episode is likely to begin in N hours”. Every definition below was frozen before any result was seen.

EVENT	observed PM2.5 >= 250 ug/m3 (CPCB "Severe" breakpoint)
WARNING	sustained >= 6 consecutive hours; events < 12 h apart are merged
HIT	the model's OWN forecast crosses the SAME threshold for the SAME duration inside its 72 h horizon
COLD HIT	an event warned by an issue time within 72 h BEFORE onset
LEAD	a hit whose earliest warning was issued while observed concentrations were still BELOW the threshold
	onset minus the EARLIEST qualifying warning

Model	POD	COLD POD	Cold lead	False alarms	Alert burden
persistence	77%	0 / 13	—	51	30%
climatology	0%	0 / 13	—	0	0%
lgbm-v1	62%	5 / 13	44 h	64	41%
lgbm-l2	69%	6 / 13	56 h	72	46%
lgbm-q90	92%	9 / 13	68 h	128	87%

8.1 Two findings that change the conclusions

Climatology cannot warn at all. Its ceiling is 247 µg/m³, below the 250 threshold, so POD is 0/13. The baseline that beat our model on MAE in 2024 is operationally worthless. The choice of metric completely inverts the ranking.

Persistence's 77% is bookkeeping. COLD POD 0/13 — it never once anticipated an episode from clean air; every “hit” was an episode already in progress. Without the cold-start split we would have reported persistence as a strong warning baseline and been wrong.

Genuine anticipation therefore belongs only to the learned models, and q90 has by far the most: **69% of severe episodes called before they began, with a median 68 hours of lead.**

8.2 The cost, stated plainly

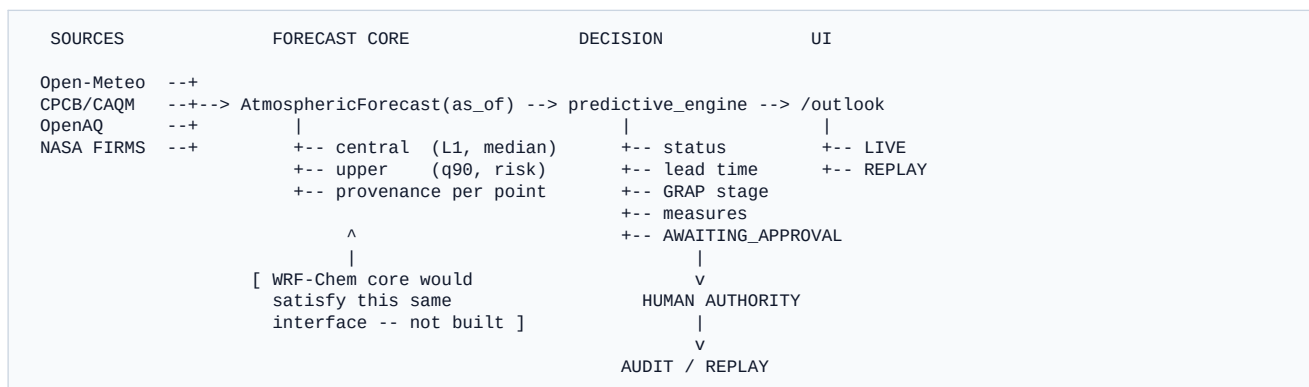
q90 sits in a warning state 87% of issue times. A system that is nearly always warning is not warning. This is the same precaution/false-alarm trade-off the ventilation threshold already exposes as balanced / precautionary operating points — the forecast objective is another operating point on that curve, and the choice belongs to the authority, not to us.

8.3 The microscope — 16–21 Nov 2024, peak 1000 µg/m³

Model	Warning
persistence	68 h before onset — but from already-severe air
climatology	NO WARNING
lgbm-v1	62 h before onset
lgbm-q90	68 h before onset

lgbm-v1 under-forecast this episode's magnitude catastrophically (bias −318 µg/m³) and still warned 62 hours ahead. **Magnitude error and warning value are different objectives** — which is precisely why AREE's claim is lead time rather than concentration.

9. Architecture as built



9.1 The one contract

Everything hangs off `AtmosphericForecast(as_of)`. The decision layer, the API and the dashboard are written against that contract, not against `LightGBM`. This is the boundary a WRF-Chem core would later satisfy, and it is why the coupled model can be substituted without rewriting anything above it.

9.2 Live and replay are the same code path

at becomes `as_of` and is threaded through every downstream call unchanged. The route never calls `datetime.now()` to fill a gap, because that is exactly how a replay quietly turns into live data. There is no separate demo mode to maintain.

9.3 Leakage is prevented structurally, not by discipline

Persisted models are named with the last date they were allowed to see, and the forecast loads the newest model whose `train_end` is at or before `as_of`. A replay of 16 Nov 2024 **cannot** load a model trained afterwards — the file that would allow it is never a candidate. Verified: a 2021 replay is refused outright rather than silently served.

9.4 Two engines, one contract

Pathway ships Linux/macOS wheels only. Rather than let three of four views die on Windows, a direct engine runs the same GRAP state machine, persistence, hysteresis and causal attribution without the streaming runtime. Both publish an identical `latest_state`, enforced by a contract test that parses both modules. The ventilation and outlook routes depend on neither, so a demonstration cannot be killed by one import failing.

9.5 The warning rule exists once

The threshold, the 6-hour persistence and the run detector live in the decision engine, and the scoring script **imports them** rather than keeping a copy. The rule running in production is provably the same function Experiment D validated. After the refactor, D was re-run and every number was identical.

10. The delivered vertical slice

One endpoint composes the whole chain. Real output from the replay anchor, served over HTTP and rendered in the browser:

```
GET /api/aree/outlook?at=2024-11-02T06:00:00Z

OBSERVATION  129 ug/m3 · Very Poor · source: legacy

ATMOSPHERE   ventilation 3695 -> 28 m2/s   falling  52 of 72 h below 466
              PBLH      1785 -> 20 m     falling
              wind       2.1 -> 0.2 m/s   falling
              collapse onset 12:00 UTC · 52 sustained h · window 6.0 h

PLUME        influence 8.79 · 1083 detections/24h · 7058 FRP · NASA FIRMS

RISK         PREDICTIVE_WARNING
              crossing 03 Nov 00:00 · lead 18 h · trigger upper_tail_q90
              central 208 · upper 305 · sustained 53 h

DECISION     triggered TRUE · priority HIGH · window 6.0 h
              AWAITING_APPROVAL · 5 measures · CAQM / DPCC

PROVENANCE   REPLAY -- reconstructed from data available as of
              2024-11-02 06:00 UTC · models central__20241101 / upper__20241101
```

Note the **two clocks**, which are genuinely different things: a 6-hour intervention window (ventilation collapses at 12:00) and an 18-hour warning lead (severe PM2.5 crossing at 00:00). The atmosphere shuts down first; the pollution follows.

10.1 Three states, driven by the backend

Status	Meaning	Observed in
MONITOR	No episode under way and none forecast	Live, September
PREDICTIVE_WARNING	No severe episode yet, but the upper tail says one is coming	51 of 78 issue times in Nov 2024
SEVERE_EPISODE_UNDERWAY	Observed concentrations already in the CPCB Severe band	27 of 78 issue times in Nov 2024

The UI branches on `status`, never on `forecast_risk` — during an ongoing episode the forecast risk is continuation information, not a new warning. No threshold comparison, risk computation or GRAP branch exists anywhere in the frontend.

10.2 Live operation

The current hour is never observed: CPCB and CAQM publish hourly and arrive 40–100 minutes late. A forecast anchored to the wall clock therefore always fails, which is the wrong question. Live forecasts anchor to the latest complete observation and say so — *“LIVE — issued 08:00 UTC, anchored to the latest observation at 07:00 UTC”* — bounded to 6 hours so it cannot silently serve yesterday.

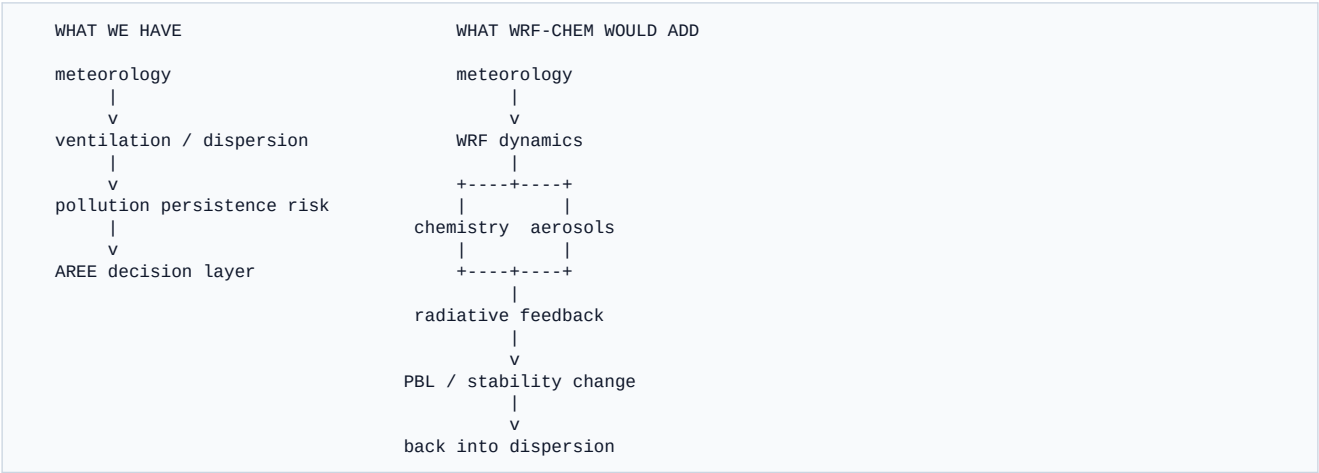
11. The route to WRF-Chem

PS 26082 asks for coupled chemistry–meteorology modelling. We have **not** built WRF-Chem, and this document does not claim otherwise. What exists is the system around that capability, plus a defined interface for the coupled core to plug into.

11.1 The honest positioning

“AREE is an operational decision layer built around a coupled atmospheric forecasting architecture. The prototype demonstrates the complete forecasting-to-action pipeline with an empirical forecast core and a defined WRF-Chem coupling interface for the full chemistry–meteorology simulation.”

What we have today is **one direction**: forecast meteorology drives a dispersion-capacity estimate, which drives a persistence risk. We do **not** have the return path in which aerosol loading modifies radiation, stability and boundary-layer depth. Claiming two-way coupling would be false, and our own λ experiment is the evidence that we do not have it.



11.2 Why we do not start by installing it

WRF-Chem is not a Python package called from FastAPI. It requires compilation of WRF and WRF-Chem, meteorological initial and boundary conditions, an emissions inventory, a chemical mechanism, aerosol configuration, domain and grid definition, preprocessing, and typically HPC-scale compute with multi-hour runtimes. Making the prototype depend on it running on a Windows development machine would put the entire demonstration at risk. **The MVP must remain runnable.**

11.3 The integration plan

Stage	Work	Gate
A. Interface	Formalise AtmosphericForecast as the coupling boundary; adapt to existing forecast output	Existing forecast output
B. Domain	Define NCR domain, grid, vertical levels; source initial/boundary conditions	Single short run completes
C. Emissions	Inventory + FIRMS fire emissions injection	Mass balance sanity checks pass
D. Chemistry	Select mechanism and aerosol scheme	Species output physically plausible
E. Validation	Score WRF-Chem output through the SAME baselines and the SAME persistence function	Beats persistence function
F. Ablation	Empirical core vs WRF-Chem vs hybrid	Quantified contribution of coupling

Stage E is the one that matters. A coupled model that cannot beat climatology on warning skill has not earned its place in the pipeline, however sophisticated its physics. The evaluation protocol is already built and frozen, so WRF-Chem can be scored the day it produces output — which is a considerable advantage over starting the validation question from scratch.

11.4 The nearer-term scientific path

Experiment B showed meteorological inputs cost only 0–3 MAE, so the coupling layer is **not** where the current error lives. The evidence points instead at the target and the objective. In priority order: rebuild the target from the accumulating multi-station capture; adopt the two-output forecast in production; re-run the frozen protocol against the new target; and only then ask whether the residual error justifies coupled physics.

12. Appendix

12.1 Reproducing everything

```
python backfill.py probe coverage before ingesting
python backfill.py met --start ... --end ... ERA5 meteorology
python backfill.py met-recent last ~92 d, incl. pressure levels
python backfill.py import-research 29,953 h historical NCR PM2.5
python backfill.py fires --start ... --end ... FIRMS detections
python backfill.py derive computed features
python backfill.py baseline / score persistence + climatology
python train_lgbm.py walk-forward lgbm-v1
python experiment_b.py operational skill with real NWP
python diagnose.py c0 / c1 / c2 target integrity + failure autopsy
python objective_experiment.py L1 / L2 / log / q90
python warning_skill.py Experiment D
python audit_history.py sensor-level availability audit
python train_forecast.py --train-end ... persist central + upper models
python capture.py bootstrap / once / loop / status
python target.py build / report / spread
```

12.2 Known gaps, stated

Gap	Consequence
3 of 4 evaluation folds rest on a single monitor	Cross-year comparisons are suspect; fixed only by accumulating multi-station data
ERA5 archive serves no pressure levels	Inversion and lapse rate are NULL historically; available for the recent ~92 days only
No forecast PBLH in the previous-runs archive	Experiment B is an interval, not a point estimate
Retroactive multi-station rebuild impossible	Today's OpenAQ sensors post-date the evaluation winters (1–3 of 95 span them)
q90 alert burden 87%	Not yet operationally economical; a calibration decision, not a plumbing one
49–72 h horizon ties climatology	72-hour skill cannot be claimed on the current evidence
WRF-Chem not built	Interface defined; see section 11

12.3 Vocabulary discipline

Never quote ~23 ms near the word “forecasting” — it is decision-layer latency, and an NWP scientist reads it as a category error. Never present λ as a working index. Never call the upper tail a prediction. Never claim WRF-Chem or two-way coupling. Never say WAQI is fully live. We did not invent the ventilation coefficient; the operational chain built on it is ours.

The system proposes. The authority disposes. Every number carries its provenance.